

EXPORT CLASSIFICATION INFRASTRUCTURE

Substrata — Export classification infrastructure for advanced hardware.

Upload a datasheet. Get extracted specifications, candidate ECCNs, citations, uncertainty flags, and a review-ready memo for human approval.

- Working product
- Evidence-backed recommendations
- Auditable human decisions

A RECURRING, HIGH-STAKES BOTTLENECK

Classification is **slow, fragmented**, and dependent on scarce expertise.

01 Facts are scattered

Datasheets, engineering notes, old memos, spreadsheets, and email live in separate systems.

02 The work repeats

Reviewers reconstruct product form, components, performance, interfaces, and security for each new product or variant.

03 The reasoning is hard to audit

Missing facts, conflicting evidence, prior decisions, and reviewer judgment are rarely preserved in one traceable record.

TODAY'S WORKFLOW

A defensible decision is assembled by hand.

The bottleneck is not only finding a possible ECCN. It is producing the evidence package a qualified reviewer can inspect and approve.

Technical source files

Control-list research

Company precedents

Reviewer rationale

When product cycles speed up, compliance teams must choose between slower releases and review records that are harder to defend.

A REVIEW WORKUP—NOT A BLACK-BOX ANSWER

Upload a datasheet. Receive a cited classification memo draft.

Extracted specifications

Product form, architecture, interfaces, memory, security, and performance.

Candidate ECCNs

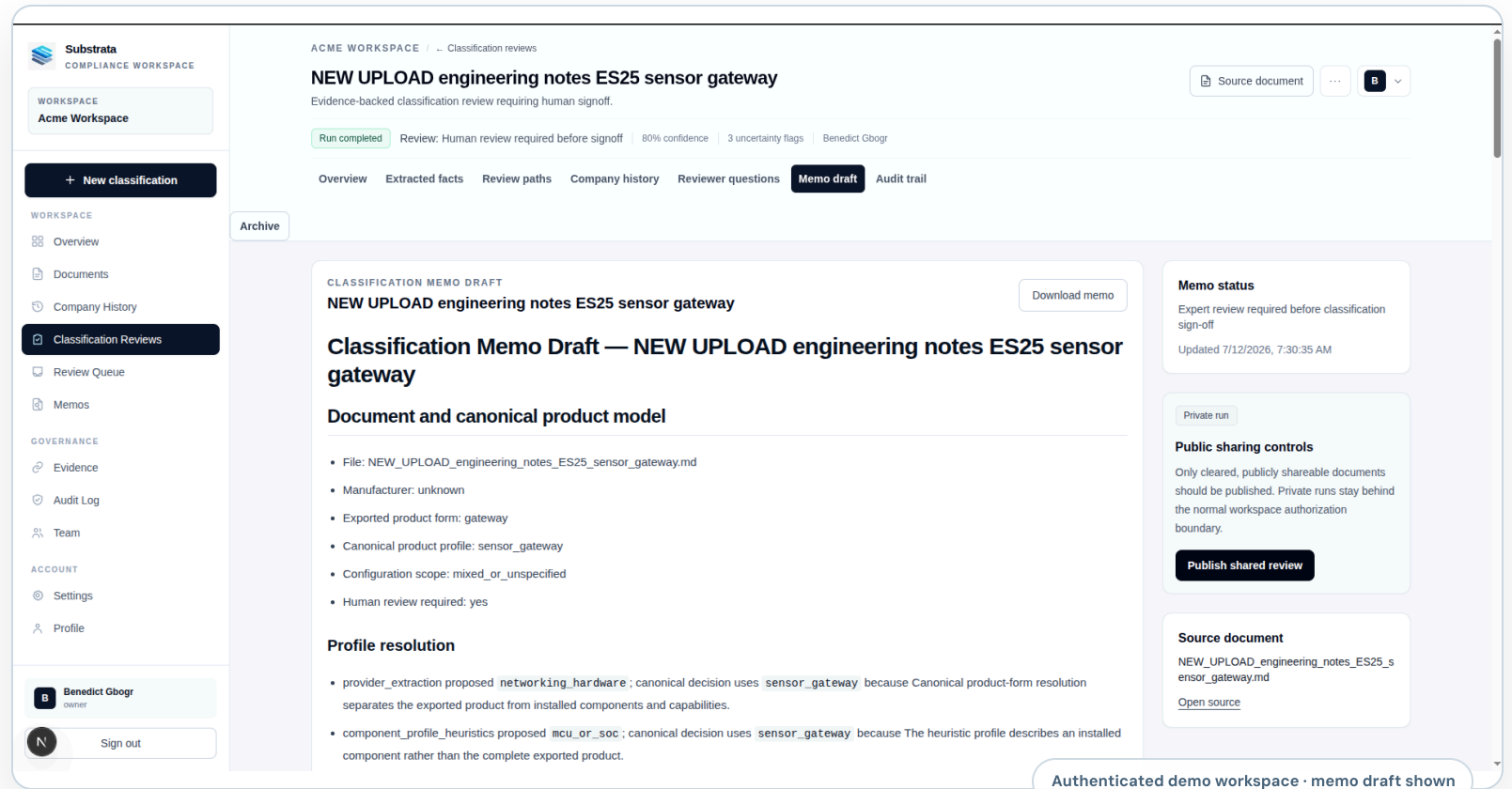
Recommended review paths with why they may or may not apply.

Citations + uncertainty

Source evidence, missing facts, conflicts, and reviewer questions.

Human-review-ready memo

A draft tied to its evidence and audit trail; human signoff remains mandatory.

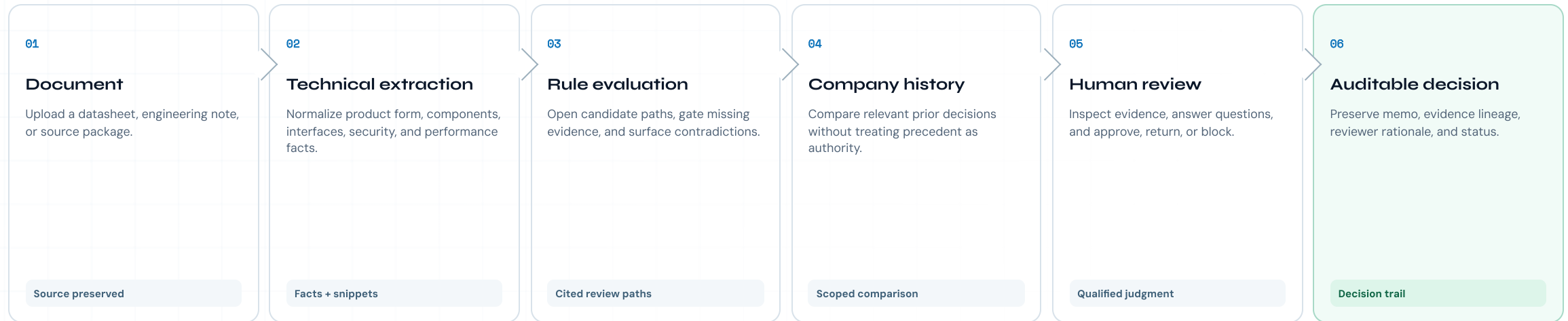


The screenshot displays the Substrata Compliance Workspace interface. The left sidebar shows a navigation menu with sections: WORKSPACE (Overview, Documents, Company History, Classification Reviews, Review Queue, Memos), GOVERNANCE (Evidence, Audit Log, Team), ACCOUNT (Settings, Profile), and a user profile for Benedict Gboggr. The main content area is titled 'ACME WORKSPACE / Classification reviews' and shows a 'NEW UPLOAD engineering notes ES25 sensor gateway' review. The review status is 'Run completed' with a 'Review: Human review required before signoff' and '80% confidence'. The 'Memo draft' tab is active, showing a 'CLASSIFICATION MEMO DRAFT' for the same title. The memo content includes 'Document and canonical product model' with a list of extracted facts (File, Manufacturer, Exported product form, Canonical product profile, Configuration scope, Human review required) and 'Profile resolution' with two entries explaining canonical decisions. On the right, there are sidebars for 'Memo status' (Expert review required, Updated 7/12/2026, 7:30:35 AM), 'Public sharing controls' (Private run, Publish shared review button), and 'Source document' (NEW_UPLOAD_engineering_notes_ES25_s ensor_gateway.md, Open source button).

Authenticated demo workspace · memo draft shown

ONE TRACEABLE DECISION PROCESS

From source document to **auditable human decision**.

**Designed for accountable review**

Substrata recommends review paths and drafts the workup. It does not replace compliance professionals, counsel, regulators, or final human decision-making.

DISCOVERY IS SHAPING A FUNCTIONING PRODUCT

Early Validation and Product Progress

CUSTOMER DISCOVERY FUNNEL

~200

Targeted outreach emails sent

Trade compliance, export controls, semiconductors, advanced hardware, and related fields.

~10

Active industry discussions

Relevant professionals helping validate the problem and refine the product.

Next

Potential design partners being identified

Discussions may lead to design-partner or adviser relationships; none are presented as formal commitments.

Practitioner input already changed the product. Feedback from Anna Goncz directly influenced the company-history and precedent-based classification workflow. She is a potential design partner or industry adviser in discussion, not a confirmed partner.

Functioning product workflow

Substrata already supports the complete review workup:

Technical-spec extraction

ECCN candidates

Supporting citations

Uncertainty flags

Company-history comparison

Audit trail

Human review

Memo generation

72 / 72

Lifecycle integration

18 / 18

API regression

32 / 32

Worker tests

What the discussions are doing

Validating the problem, refining workflow requirements, testing the value of precedent-aware review, and identifying potential design partners.

THE ANSWER IS NOT THE PRODUCT

Companies need a **traceable decision process**, not only an AI answer.

“A plausible ECCN without evidence lineage, uncertainty, precedent context, and accountable review does not solve the compliance workflow.”

Substrata's product thesis, informed by product work and early reviewer feedback.

01

Product form before component signals

A gateway containing chips is not classified as if it were only the installed chip. The exported item and component roles must stay distinct.

02

Conflicts must block clean conclusions

Contradictory or missing evidence becomes an explicit uncertainty flag and reviewer task.

03

Company history is context, not authority

Prior decisions improve consistency only when provenance, similarity limits, and current evidence remain visible.

04

Human accountability is part of the system

The reviewer's decision, rationale, and evidence lineage belong in the same durable record.

THE COST OF MANUAL REVIEW IS COMPOUNDING

More restrictions. More complex products. More pressure for **fast, defensible decisions.**

01

Semiconductor restrictions are expanding

Advanced-computing and semiconductor controls create more review paths, thresholds, and configuration-specific questions for global hardware teams.

02

Products combine more functions

Compute, networking, memory, security, RF, and software capabilities increasingly coexist in one exported product.

03

Defensibility matters as much as speed

Teams need faster preparation without losing source evidence, company context, uncertainty, or qualified human approval.

The opening: turn technical-document review into reusable compliance infrastructure, beginning with datasheet-to-ECCN workups.

A FOCUSED BEACHHEAD WITH RECURRING USAGE

Start with semiconductor and specialized-hardware manufacturers.

INITIAL ANNUAL SOFTWARE OPPORTUNITY · FOUNDER ESTIMATE

Bottom-up, not a third-party TAM claim

2,000target organizations at
midpoint

×

\$40Kmidpoint initial annual contract
value

=

\$80M

annual beachhead opportunity

Sensitivity range: 1,500–3,000 target organizations × \$30K–\$75K ACV = **\$45M–\$225M**. Assumes 3–10 relevant users and roughly 100–500 classification/review events per organization per year. These are founder assumptions to validate, not verified market data.

Buyer and users

Trade-compliance leaders, export-control analysts, technical reviewers, and qualified advisers inside semiconductor and advanced-hardware companies.

Recurring trigger

New products, variants, customer requests, supplier changes, acquisitions, and updates to controls create repeated review work.

B2B SaaS model

Annual subscription priced by organization, review volume, users, and enterprise deployment requirements.

Organization

Review volume

Seats

Enterprise deployment

Expansion path

From datasheet-to-ECCN review into the decision layer for recurring trade-compliance work across advanced-hardware product families.

TYPICAL WORKFLOW COMPARISON

Substrata connects the work that existing options leave fragmented.

Capability	Spreadsheets	Consultants	Generic AI tools	Legacy trade software	Substrata
Understands technical documents	Manual	Human-led	Partial	Limited	Built in
Evidence-backed reasoning + citations	Manual	Deliverable-dependent	Inconsistent	Partial	Built in
Company classification history	Fragmented	Manual	No native record	Records, not reasoning	Native + scoped
Auditable decision lineage	Weak	Document-based	Weak	Workflow logs	Fact → decision
Explicit uncertainty + conflicts	Manual	Human-led	Often hidden	Limited	First-class
Mandatory human review	Process-dependent	Yes	Not inherent	Configurable	System boundary

Positioning: consultants bring expertise; legacy systems manage transactions; spreadsheets remain flexible; generic AI can accelerate drafting. Substrata is built around technical-document understanding, evidence lineage, company context, uncertainty handling, and accountable human review in one workflow.

FOUNDER-LED TECHNICAL EXECUTION

Two builders combining product conviction with infrastructure execution.



Benedict Gbogr

CO-FOUNDER · INFRASTRUCTURE & ENGINEERING

Founder of Jungle Grid, an execution layer for AI agents and GPU workloads. Builds Substrata's application, orchestration, worker execution, artifact, and reliability infrastructure.

Execution evidence: authenticated multi-tenant product, end-to-end classification lifecycle, integration coverage, and production-oriented deployment work in the repository.



Nathan Ulanja

CO-FOUNDER · PRODUCT & COMPLIANCE WORKFLOW

Originated Substrata's focused product direction: turn manual export-control research into cited, human-review-ready workups for semiconductor and advanced-hardware teams.

Reason for pursuing the problem: classification work is slowed by fragmented evidence, repeated technical extraction, weak precedent handling, and missing decision lineage.

Founder-led product development

The founders are building the product while validating workflow assumptions with compliance stakeholders and planning expert-reviewed validation.

WHAT \$750K UNLOCKS

Turn working validation into a **production-ready enterprise product.**

15–18
months of runway

Complete production readiness, work with trade-compliance experts, onboard design partners, and convert early validation into paid enterprise pilots.

01

Production-grade product and security

Managed storage, enterprise controls, retention, audit hardening, and operational readiness.

02

Expanded classification and rule coverage

More hardware profiles, current rule paths, stronger parsing, and clearer evidence gates.

03

Expert-reviewed validation

Trade-compliance experts review workflows, outputs, boundaries, and validation sets.

04

5–10 design partners

Semiconductor or advanced-hardware teams using the product in structured workflow sessions.

05

3–5 paid enterprise pilots

Convert validated use cases into paid, time-bound deployments with success criteria.

06

Measured product impact

Evidence that Substrata reduces review time while improving consistency, traceability, and audit readiness.

THE VISION

The compliance decision layer for every advanced-hardware product moving across borders.

Datasheet-to-ECCN review is the wedge. The durable opportunity is the system of record for evidence-backed, human-approved trade-compliance decisions.

Raising \$750K to turn Substrata into the classification system of record for advanced-hardware companies.

15–18 months
runway

Pre-seed round · No valuation cap or detailed investment terms included in this deck.

- Production readiness
- Expert validation
- Target: design partners
- Target: paid pilots